Advanced Database Systems Portfolio

MIT 8103 - Advanced Database Systems

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Case Study

APEX DIGITAL BANK

About Apex Digital Bank

Apex Digital Bank is a fintech-based digital banking platform that provides financial services to individual and corporate customers through its web and mobile channels. The platform enables customers to conveniently conduct banking activities online, including account opening, fund transfers, make deposits, monitor transactions, and manage their banking activities electronically.

Database Users

The primary users of the database include:

- Customers
- Customer Service Officers
- Operations Officers
- Database Administrators
- Compliance Officers
- Management Staff

Major Data Stored

The database stores:

- Customer information
- Bank accounts
- Employee

- Transaction records (Deposit and Withdrawals and other transactions)
- Transaction logs
- KYC records
- Debit card information
- Loan
- Branch information
- Statement

Key Operations Supported

The database supports the following operations:

- Transaction Processing: The database processes financial transactions such as deposits, withdrawals, fund transfers, bill payments, and transaction reversals. It ensures that account balances are updated accurately and that all transactions are securely recorded for audit and reporting purposes.
- Customer registration
- Account creation
- Fund transfers
- Balance enquiries
- Loan management
- KYC verification
- Account statement generation

Entity Identification: Key Entities and Descriptions

The following key entities were identified for the Apex Digital Bank database system:

- **Customer**
  Stores personal and contact information of individual and corporate customers.
- **KYC_Record**
  Stores customer verification information and compliance documentation.
- **Account**
  Stores customer account details including account type and current balance.
- **Transaction**
  Records financial transactions such as deposits, withdrawals, transfers, bill payments, and reversals.

- **Transaction_Log**
  Provides an audit trail of transaction activities performed within the system.
- **Card**
  Stores debit card information associated with customer accounts.
- **Loan**
  Stores loan application and repayment information.
- **Branch**
  Stores branch or operational office information.
- **Employee**
  Stores details of bank employees such as customer service officers, IT and operations officers, and compliance officers.